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# Safety data sheet



BASF Safety data sheet according to Regulation (EC) No. 1907/2006 as amended from time to time.
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Product: **Lutavit® E 50**
(ID no. 30040915/SDS_GEN_IE/EN)
Date of print 27.06.2025

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

## Lutavit® E 50

The substance/mixture contains nanoforms.

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: feed additive(s)

### 1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY

Contact address:
BASF Ireland DAC
Asgard House, 19-20 City Quay
Dublin, D02 K744
Ireland

Telephone: +353 21 451-7100
E-mail address: product-safety-uk-and-ireland@basf.com

### 1.4. Emergency telephone number

For products classified as hazardous in accordance with CLP:
National Poisons Information Centre, Beaumont Hospital, Dublin 9
Emergency medical information: 8am-10pm (seven days)
Tel.: 01 8092566
International emergency number:
Telephone: +49 180 2273-112

## SECTION 2: Hazards Identification

### 2.1. Classification of the substance or mixture

For the classification of the mixture the following methods have been applied: extrapolation on the concentration levels of the hazardous substances, on basis of test results and after evaluation of experts. The methodologies used are mentioned at the respective test results.

According to Regulation (EC) No 1272/2008 [CLP]

No need for classification according to GHS criteria for this product.

### 2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

The product does not require a hazard warning label in accordance with GHS criteria.

### 2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

The product is under certain conditions capable of dust explosion.
The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria. Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

## SECTION 3: Composition/Information on Ingredients

### 3.1. Substances

Not applicable

### 3.2. Mixtures

Chemical nature

Preparation based on:3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate (Content (W/W): >= 50 %), Silicon dioxide

Regulatory relevant ingredients

Silicon dioxide

Content (W/W): >= 44 % - <= 48 %
CAS Number: 7631-86-9
EC-Number: 231-545-4
REACH registration number: 01-2119379499-16

Particle characteristics of nanoform:

| | |
|---|---|
| Particle size distribution: | 2.5 - 50 nm (D50, Counted Distribution) |
| Particle Shape: | spheroidal |
| Crystallinity: | amorphous |
| Surface treatment/coating: | no |

## SECTION 4: First-Aid Measures

### 4.1. Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Rinse mouth and then drink 200-300 ml of water.

### 4.2. Most important symptoms and effects, both acute and delayed

Symptoms: (Further) symptoms and / or effects are not known so far

### 4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Symptomatic treatment (decontamination, vital functions).

## SECTION 5: Fire-Fighting Measures

### 5.1. Extinguishing media

Suitable extinguishing media:
water spray, foam, dry powder, carbon dioxide

Unsuitable extinguishing media for safety reasons:
water jet

Additional information:
Avoid whirling up the material/product because of the danger of dust explosion.

### 5.2. Special hazards arising from the substance or mixture

Endangering substances: harmful vapours, carbon oxides
Advice: Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire. Dust explosion hazard.

### 5.3. Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.
Cool endangered containers with water-spray.

## SECTION 6: Accidental Release Measures

Dust can form an explosive mixture with air.

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Information regarding personal protective measures, see section 8.
Avoid dust formation.

### 6.2. Environmental precautions

Do not discharge into drains/surface waters/groundwater.

### 6.3. Methods and material for containment and cleaning up

For small amounts: Contain with dust binding material and dispose of.
For large amounts: Sweep/shovel up.
Dispose of absorbed material in accordance with regulations. Avoid raising dust.

### 6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

## SECTION 7: Handling and Storage

### 7.1. Precautions for safe handling

Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:
Avoid dust formation. Avoid whirling up the material/product because of the danger of dust explosion. Take precautionary measures against static discharges. Avoid all sources of ignition: heat, sparks, open flame.

### 7.2. Conditions for safe storage, including any incompatibilities

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE), Polypropylene (PP)
Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

### 7.3. Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

## SECTION 8: Exposure Controls/Personal Protection

### 8.1. Control parameters

Components with occupational exposure limits

7631-86-9: Silicon dioxide
TWA value 2.4 mg/m3 (OEL (IE)), Respirable dust
TWA value 6 mg/m3 (OEL (IE)), Total inhalable dust

### 8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if dusts are formed. Particle filter with low efficiency for solid particles (e.g. EN 143 or 149, Type P1or FFP1)

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen based on level of activity and exposure.

General safety and hygiene measures

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is recommended. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift. Store work clothing separately.

## SECTION 9: Physical and Chemical Properties

### 9.1. Information on basic physical and chemical properties

| | | |
|---|---|---|
| State of matter: | solid | |
| Form: | powder | |
| Colour: | white to off-white | |
| Odour: | almost odourless | |
| Odour threshold: | not determined | |
| Melting point: | not relevant | |
| Boiling point: | not applicable | |
| Flammability: | not highly flammable | (Directive 92/69/EEC, A.10) |
| Lower explosion limit: | For solids not relevant for classification and labelling. | |
| Upper explosion limit: | For solids not relevant for classification and labelling. | |
| Flash point: | not applicable, the product is a solid | |
| Thermal decomposition: | >= 175 °C (DSC (DIN 51007)) | |
| SADT: | > 75 °C<br>Heat accumulation / Dewar 500 ml (SADT, UN-Test H.4, 28.4.4) | |
| pH value: | insoluble | |
| Viscosity, kinematic: | not applicable, the product is a solid | |
| Viscosity, dynamic: | not applicable, the product is a solid | |
| Solubility in water: | insoluble | |
| Partitioning coefficient n-octanol/water (log Kow): | not applicable for mixtures | |
| Vapour pressure: | not applicable | |
| Density: | No information is available for the absolute density. Instead the bulk density was determined as a more relevant value. | |
| Relative vapour density (air): | The product is a non-volatile solid. | |

Particle characteristics

Contains agglomerates / aggregates of nanoparticles - For particle characteristics of nanoform components refer to SDS section 3. - Potential for release of individual nanoparticles is limited.

### 9.2. Other information

**Information with regard to physical hazard classes**

Explosives

| | | |
|---|---|---|
| Explosion hazard: | Product is not explosive, however a dust explosion could result from an air / dust mixture. | |

Oxidizing properties

| | | |
|---|---|---|
| Fire promoting properties: | Based on its structural properties the product is not classified as oxidizing. | |

Flammable solids

| | | |
|---|---|---|
| Burning rate: | The material doesn't meet the criteria specified in paragraph 33.2.4.4 of UN manual of tests and criteria. | (Directive 92/69/EEC, A.10) |

Self-heating substances and mixtures

| | | |
|---|---|---|
| Self heating ability: | It is not a substance capable of spontaneous heating according to UN transport regulations class 4.2. | (UN Test N.4 (self heating substances)) |

Corrosion to metals

Corrosive effects to metal are not anticipated.

**Other safety characteristics**

| | | |
|---|---|---|
| Minimum ignition energy: | 1 - 3 mJ<br>(20 °C)<br>Inductivity: 1 mH<br>Grain size distribution: < 63 µm<br>The product is capable of dust explosion. | (VDI 2263, sheet 1, 2.5) |
| Bulk density: | approx. 450 - 600 kg/m3 | |
| SAPT-Temperature: | Study scientifically not justified. | |
| Evaporation rate: | negligible | |

## SECTION 10: Stability and Reactivity

### 10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: Corrosive effects to metal are not anticipated.

### 10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated.

### 10.3. Possibility of hazardous reactions

Dust can form an explosive mixture with air.

### 10.4. Conditions to avoid

Avoid dust formation. Avoid electro-static charge. See SDS section 7 - Handling and storage.

### 10.5. Incompatible materials

Substances to avoid:
Alkalines, atmospheric moisture

### 10.6. Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

## SECTION 11: Toxicological Information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: 3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*

*Experimental/calculated data:*
*LD50 rat (oral): > 10,000 mg/kg (BASF-Test)*

*Information on: Silicon dioxide*

*Experimental/calculated data:*
*LD50 rat (oral): > 5,000 mg/kg (OECD Guideline 401)*
---

Irritation

Assessment of irritating effects:
Not irritating to the eyes. Not irritating to the skin. The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: 3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Experimental/calculated data:*
*Skin corrosion/irritation*
*rabbit: non-irritant (OECD Guideline 404)*
---

*Information on: 3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Experimental/calculated data:*
*Serious eye damage/irritation*
*rabbit: non-irritant (OECD Guideline 405)*
---

Respiratory/Skin sensitization

Assessment of sensitization:
Skin sensitizing effects were not observed in animal studies.

*Information on: 3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Assessment of sensitization:*
*Skin sensitizing effects were not observed in animal studies.*

*Information on: Silicon dioxide*
*Assessment of sensitization:*
*Skin sensitizing effects were not observed in animal studies.*
---

Germ cell mutagenicity

Assessment of mutagenicity:
Based on available data, the classification criteria are not met.

*Information on: 3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Assessment of mutagenicity:*
*No mutagenic effect was found in various tests with bacteria and mammals.*

*Information on: Silicon dioxide*
*Assessment of mutagenicity:*
*The substance was not mutagenic in bacteria. The substance was not mutagenic in a test with mammals.*
---

Carcinogenicity

Assessment of carcinogenicity:

In long-term animal studies in which the substance was given in high doses by feed, a carcinogenic effect was not observed. The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: 3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Assessment of carcinogenicity:*
*In long-term animal studies in which the substance was given in high doses by feed, a carcinogenic effect was not observed.*

*Information on: Silicon dioxide*
*Assessment of carcinogenicity:*
*Not classified, due to lack of data.*
---

Reproductive toxicity

Assessment of reproduction toxicity:
Based on available data, the classification criteria are not met.

*Information on: 3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Assessment of reproduction toxicity:*
*The results of animal studies gave no indication of a fertility impairing effect.*

*Information on: Silicon dioxide*
*Assessment of reproduction toxicity:*
*Not classified, due to lack of data.*
---

Developmental toxicity

Assessment of teratogenicity:
Based on the ingredients, there is no suspicion of a teratogenic effect.

*Information on: 3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Assessment of teratogenicity:*
*No indications of a developmental toxic / teratogenic effect were seen in animal studies.*

*Information on: Silicon dioxide*
*Assessment of teratogenicity:*
*Not classified, due to lack of data.*
---

Specific target organ toxicity (single exposure)

Remarks: Based on available data, the classification criteria are not met. The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on available data, the classification criteria are not met.

*Information on: 3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Assessment of repeated dose toxicity:*
*Repeated oral uptake of the substance did not cause substance-related effects.*

*Information on: Silicon dioxide*
*Assessment of repeated dose toxicity:*
*Not classified, due to lack of data.*
---

Aspiration hazard

No aspiration hazard expected.

Interactive effects

No data available.

## 11.2. Information on other hazards

Endocrine disrupting properties

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

Other information

Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

## SECTION 12: Ecological Information

### 12.1. Toxicity

Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. The product has not been tested. The statement has been derived from the properties of the individual components.
---

*Information on:3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Toxicity to fish:*
*LC50 (96 h) > 11 mg/l, Oncorhynchus mykiss (OECD Guideline 203, static)*
*The statement of the toxic effect relates to the analytically determined concentration. No toxic effects occur within the range of solubility.*

*Information on:Silicon dioxide*
*Toxicity to fish:*
*LC50 (96 h) > 10,000 mg/l, Brachydanio rerio (OECD 203; ISO 7346; 84/449/EEC, C.1)*
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*Information on:3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Aquatic invertebrates:*
*EC50 (48 h) > 20.6 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)*
*The statement of the toxic effect relates to the analytically determined concentration. No toxic effects occur within the range of solubility.*

*Information on:Silicon dioxide*
*Aquatic invertebrates:*
*EC50 (24 h) > 1,000 mg/l, Daphnia magna (OECD Guideline 202, part 1)*
---

*Information on:3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Microorganisms/Effect on activated sludge:*
*EC20 (30 min) > 927 mg/l, activated sludge, domestic (DIN EN ISO 8192, aquatic)*
*The details of the toxic effect relate to the nominal concentration.*

*Information on:Silicon dioxide*
*Microorganisms/Effect on activated sludge:*
*EC50 (3 h) > 2,500 mg/l (OECD Guideline 209)*
---

### 12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):

The product is not very soluble in water and can thus be removed from water mechanically in suitable effluent treatment plants.

*Information on:3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Assessment biodegradation and elimination (H2O):*
*Moderately/partially biodegradable. Not readily biodegradable (by OECD criteria). The product is virtually insoluble in water and can thus be separated from water mechanically in suitable effluent treatment plants.*

*Information on:Silicon dioxide*
*Assessment biodegradation and elimination (H2O):*
*Inorganic product which cannot be eliminated from water by biological purification processes.*
---

## 12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

*Information on:3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Assessment bioaccumulation potential:*
*Accumulation in organisms is not to be expected.*

*Information on:Silicon dioxide*
*Assessment bioaccumulation potential:*
*The product will not be readily bioavailable due to its consistency and insolubility in water.*
---

## 12.4. Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: Adsorption to solid soil phase is expected.

*Information on:3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate*
*Assessment transport between environmental compartments:*
*Volatility: The substance will slowly evaporate into the atmosphere from the water surface.*
*Adsorption in soil: Adsorption to solid soil phase is expected.*

*Information on:Silicon dioxide*
*Assessment transport between environmental compartments:*
*Adsorption in soil: Study scientifically not justified.*
---

### 12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria. Self classification

### 12.6. Endocrine disrupting properties

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

### 12.7. Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

### 12.8. Additional information

Add. remarks environm. fate & pathway:
The product has not been tested. The statements on environmental fate and pathway have been derived from the properties of the individual components.

## SECTION 13: Disposal Considerations

### 13.1. Waste treatment methods

Observe national and local legal requirements.

Contaminated packaging:
Uncontaminated packaging can be re-used.
Packs that cannot be cleaned should be disposed of in the same manner as the contents.

## SECTION 14: Transport Information

**Land transport**

ADR

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |

| | |
|---|---|
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

RID

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user: | None known |

Transport in inland waterway vessel
Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Air transport**

IATA/ICAO

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

### 14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

### 14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

### 14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

### 14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

### 14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

### 14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

### 14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

## SECTION 15: Regulatory Information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
Listed in above regulation: no

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

### 15.2. Chemical Safety Assessment

Advice on product handling can be found in sections 7 and 8 of this safety data sheet.

## SECTION 16: Other Information

Any other intended applications should be discussed with the manufacturer. Corresponding occupational protection measurements must be followed.

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.